TCP SYN Flood (Denial of Service) — Security Assessment Report

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1. Executive Summary

This report documents a controlled TCP SYN flood denial-of-service attack conducted against a Windows target host within an isolated virtual laboratory, together with the detection and analysis performed from the defensive side. The objective was to reproduce a volumetric denial-of-service condition and to establish the observable characteristics that distinguish it from reconnaissance activity such as a port scan.

A SYN flood was executed from a Linux host against port 80 of the target using hping3. The traffic was captured on the target with Wireshark and analysed. The attack was positively identified through its characteristic signature: a single source host transmitting a sustained, very high volume of half-open TCP connection attempts to a single destination port, with no completed handshakes. The activity corresponds to MITRE ATT&CK techniques T1498 (Network Denial of Service) and T1499.001 (Endpoint Denial of Service: OS Exhaustion Flood).

The exercise confirmed that a SYN flood is readily detectable through packet rate analysis and host connection-state inspection, and it provided a direct contrast with the port scan documented separately, clarifying how the two SYN-based activities differ in intent and signature.

2. Objective

The objectives of this assessment were:

- To perform a TCP SYN flood against a single service port on a controlled target host.
- To capture and analyse the associated network traffic from the defender's perspective.
- To document the observable indicators of a volumetric denial-of-service condition.
- To define appropriate detection and mitigation measures.

3. Scope and Test Environment

All activity was confined to an isolated VMware host-only network with no connectivity to production systems or the internet. The scope was limited to the two hosts listed below.

Attacker host: Linux virtual machine, IP address 192.168.213.132, hardware address 00:0c:29:03:50:b4 Target host: Windows virtual machine, IP address 192.168.213.133, hardware address 00:0c:29:1c:9e:b5 Targeted service port: TCP 80 Network segment: 192.168.213.0/24 (isolated, host-only) Monitoring: Wireshark packet capture running on the target host

No systems outside this segment were in scope, and no production or third-party assets were affected at any point. To preserve the stability of the environment, the attack was limited to a short duration of approximately fifteen seconds.

4. Tools Used

hping3 was used on the attacker host to generate the SYN flood. Wireshark was used on the target host to capture and analyse the resulting traffic. Standard Windows command-line and monitoring utilities (netstat, Resource Monitor, Task Manager) were used to observe the effect on the target.

5. Methodology

The assessment followed a controlled attack-and-detection methodology. Network connectivity between the hosts was confirmed, and a packet capture was started on the target. A command prompt was prepared on the target in advance to permit live inspection of connection states during the attack, as the most relevant host-based evidence exists only while the flood is active. The attack was then executed for a limited period, after which the traffic and host state were analysed.

6. Execution

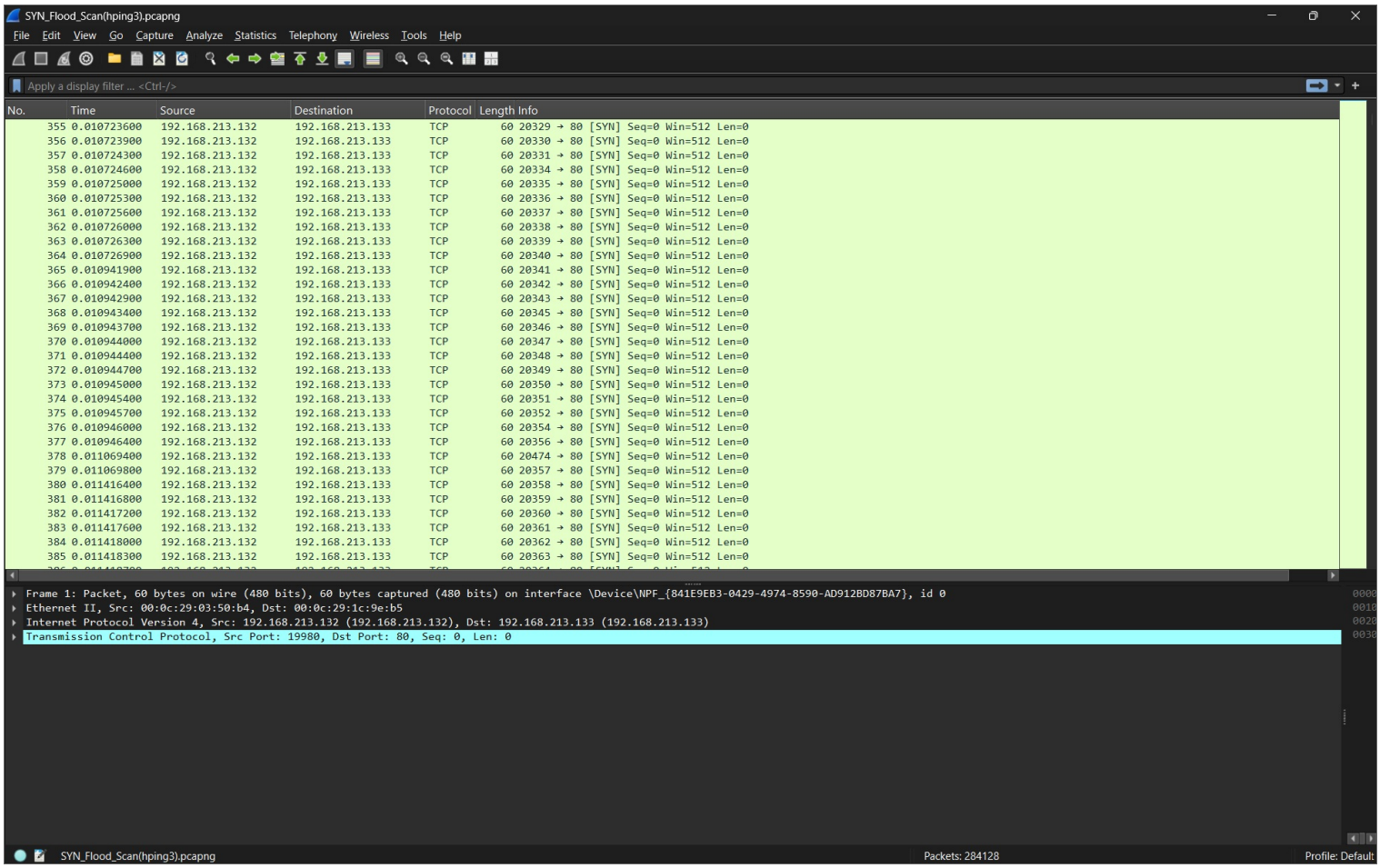
The attack was performed using hping3 in flood mode with the SYN flag set:

```
sudo hping3 -S --flood -p 80 192.168.213.133
```

The -S option sets the TCP SYN flag on each transmitted packet, the --flood option transmits packets at maximum rate without waiting for responses, and the -p 80 option directs all traffic to port 80 on the target. The attack exploits the TCP connection-establishment process: each SYN causes the target to allocate resources for a half-open connection and to respond with a SYN/ACK while awaiting completion of the handshake. Because the attacker never transmits the final ACK, these half-open connections accumulate in the target's connection backlog queue. When the queue is exhausted, the target is unable to accept further connections, including legitimate ones, resulting in denial of service.

Source-address randomisation (the --rand-source option) was deliberately not used. Retaining the attacker's genuine source address allowed the traffic to be reliably attributed to a single host during analysis, which was consistent with the detection-focused objective of the exercise.

7. Observations and Evidence



Analysis of the captured traffic produced the following findings.

All observed packets originated from the single source address 192.168.213.132 and were directed to the single destination address 192.168.213.133 on port 80. Unlike the port scan, in which the destination port varied continuously, the destination remained constant throughout, indicating a concentrated attack against a single service.

The volume of traffic was substantial, with the capture recording in excess of 284,000 packets over the short duration of the attack. Packet timestamps were separated by only microseconds (for example 0.010723, 0.010724 and 0.010725 seconds), corresponding to a sustained rate of several thousand packets per second, far exceeding any rate associated with legitimate client activity.

Each captured packet was uniform in structure: a single SYN packet with sequence number 0, a window size of 512, no payload, and a length of 60 bytes. The source port incremented in near-sequential fashion across successive packets (for example 20329, 20330, 20331, 20334 and 20335), consistent with hping3 assigning a new source port to each transmitted packet.

Examination of the Ethernet layer confirmed that all traffic passed between the attacker's hardware address (00:0c:29:03:50:b4) and the target's hardware address (00:0c:29:1c:9e:b5), both VMware interfaces, establishing attribution at the link layer.

The following Wireshark display filter was applied to isolate the relevant traffic:

```
tcp.flags.syn == 1 && tcp.flags.ack == 0 && tcp.port == 80
```

Supporting statistical analysis was performed within Wireshark. The I/O Graph showed a sharp increase in packet rate that remained sustained for the full duration of the attack, in contrast to the brief bursts characteristic of a scan. The Conversations view showed a single conversation carrying an extremely high packet count with minimal data per packet, consistent with a flood of empty SYN packets.

8. Detection and Analysis

The defining signature of the activity is a single source host transmitting a sustained, very high volume of lone SYN packets to a single destination port, accompanied by a large number of half-open connections on the target.

Host-based detection was confirmed using netstat. Under normal conditions the number of connections in the SYN_RECEIVED state is negligible; during the attack this figure rose sharply. The count can be obtained directly as follows:

```
netstat -an | find /c "SYN_RECEIVED"
```

The elevated SYN_RECEIVED count represents the clearest host-level indicator of an active SYN flood. Resource Monitor and Task Manager additionally showed a sustained increase in inbound network activity, and the target exhibited degraded responsiveness during the attack, consistent with the denial-of-service condition. In an enterprise environment, this activity would typically be detected through packet inspection by an intrusion detection system such as Suricata, with correlation and alerting handled by a SIEM platform such as Wazuh.

9. Comparison with Port Scan Activity

Although both the port scan and the SYN flood are constructed from TCP SYN packets, they differ fundamentally in intent and signature. A port scan contacts many distinct destination ports at a moderate rate and relies on the responses in order to enumerate open services. A SYN flood concentrates on a single destination port at the maximum achievable rate and disregards responses, its purpose being resource exhaustion rather than information gathering. Two indicators therefore allow rapid differentiation during analysis: the number of destination ports contacted, and the rate of packet transmission. A wide range of ports at a moderate rate indicates reconnaissance, whereas a single port at a very high sustained rate indicates a denial-of-service attempt.

10. Impact Assessment

A successful SYN flood degrades or denies the availability of the targeted service by exhausting the target's connection backlog, preventing legitimate connections from being established. In this exercise the target exhibited measurable performance degradation during the attack. In a production context, an attack of sufficient volume against an exposed service could render that service unavailable to legitimate users for the duration of the attack. The impact is therefore assessed as significant with respect to availability, though the attack does not compromise the confidentiality or integrity of the target.

11. Recommendations

The following measures are recommended to reduce susceptibility to, and improve detection of, SYN flood activity:

- Enable SYN cookies, which allow the operating system to validate incoming connections without allocating backlog resources until a connection is confirmed, thereby neutralising the resource-exhaustion mechanism.
- Apply firewall rate-limiting to restrict the number of new SYN packets accepted from a single source within a given interval.
- Deploy an intrusion prevention system with SYN-rate threshold rules to detect and block flood activity automatically.
- For internet-facing services, employ upstream distributed denial-of-service protection to absorb high-volume traffic before it reaches the host.

12. Conclusion

The TCP SYN flood was successfully executed and positively identified through network traffic and host connection-state analysis. The activity was characterised by a single source host transmitting a sustained, very high volume of half-open connection attempts to a single destination port, with no completed handshakes and a corresponding accumulation of half-open connections on the target. When compared with the port scan documented separately, the exercise clearly demonstrated the distinction between reconnaissance and denial-of-service activity, and reinforced the detection methods applicable to each.

This assessment was conducted in an isolated laboratory environment on systems owned by the author, for educational and defensive-research purposes only.